

# Jiawei Lian, Curriculum Vitae

✉ [jiawei.lian@connect.polyu.hk](mailto:jiawei.lian@connect.polyu.hk); [lianjiawei@mail.nwpu.edu.cn](mailto:lianjiawei@mail.nwpu.edu.cn)  
☎ +86-15978793013, +852-60891479     [Google Scholar](#)  
 [Research Gate](#)     [Github](#)     [Homepage](#)



-**Birth Date:** 01/1997

-**Status:** Final-year PhD student, PolyU and NWPU joint training leading to dual PhD

-**Research Interest:** AI Safety, Large language model, Computer vision, Remote sensing

-**Postal Address:** Room 505, Building CF, The Hong Kong Polytechnic University, Hung Hom, Kowloon, Hong Kong

## Education

- 01/2024 – Present     Ph.D., Artificial Intelligence and Signal Processing, Advisor: [Lap-Pui Chau](#)  
The Hong Kong Polytechnic University (PolyU)  
Thesis: *Towards Exploring the Adversarial Vulnerability of Deep Learning Models*
- 04/2022 – 06/2026     Ph.D., Information and Communication Engineering, Advisor: [Shaohui Mei](#)  
Northwestern Polytechnical University (NWPU)  
Thesis: *Towards Exploring the Adversarial Vulnerability of Deep Learning Models*
- 09/2019 – 03/2022     M.E., Control Engineering, Advisor: [Junhong He](#)  
Northwestern Polytechnical University  
Thesis: *Research on Rapid Detection of Surface Defects Based on Deep Learning*
- 09/2015 – 07/2019     B.E., Automation, Advisor: Junlin Zhu  
JiangXi University of Science and Technology  
Thesis: *Design of 4R Joint Robot Control System based on PROFIBUS and T-CPU*

## Research Publications

### Journal Articles

1. **Jiawei Lian**, Jianhong Pan, Lefan Wang, Yi Wang, Xiaofei Wang, Yingjie Lu, Shaohui Mei, Lap-Pui Chau. "Revealing the Intrinsic Ethical Vulnerability of Aligned Large Language Models." *Nature Communications (NC)*, 2026. [\[Link\]](#)
2. **Jiawei Lian**, Jianhong Pan, Lefan Wang, Yi Wang, Tairan Huang, Shaohui Mei, Lap-Pui Chau. "LLM-Agnostic Semantic Representation Attack." *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2026. [\[Link\]](#)
3. **Jiawei Lian**, Shaohui Mei, Xiaofei Wang, Yi Wang, Lefan Wang, Yingjie Lu, Mingyang Ma, Lap-Pui Chau. "Blind Object Detectors via Transferable Background Adversarial Attack." *Neural Networks (NN)*, 2026. [\[Link\]](#)
4. **Jiawei Lian**, Jianhong Pan, Lefan Wang, Yi Wang, Shaohui Mei, Lap-Pui Chau. "PADetBench: Towards Benchmarking Physical Attacks against Object Detection." *Knowledge-Based Systems (KBS)*, 2025. [\[Link\]](#)
5. Shaohui Mei (PhD Advisor), **Jiawei Lian**, Xiaofei Wang, Yuru Su, Mingyang Ma, Lap-Pui Chau. "A Comprehensive Study on the Robustness of Image Classification and Object Detection in Remote Sensing: Surveying and Benchmarking." *Journal of Remote Sensing (JRS)*, 2024. [\[Link\]](#)
6. **Jiawei Lian**, Xiaofei Wang, Yuru Su, Mingyang Ma, Shaohui Mei. "CBA: Contextual Background Attack against Optical Aerial Detection in the Physical World." *IEEE Transactions on Geoscience and Remote Sensing (TGRS)*, 2023. [\[Link\]](#)
7. **Jiawei Lian**, Shaohui Mei, Shun Zhang, Mingyang Ma. "Benchmarking Adversarial Patch against Aerial Detection." *IEEE Transactions on Geoscience and Remote Sensing (TGRS)*, 2022. [\[Link\]](#)

### Conference Papers

1. **Jiawei Lian**, Jianhong Pan, Lefan Wang, Yi Wang, Shaohui Mei, Lap-Pui Chau. "Semantic Representation Attack against Aligned Large Language Models." *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. [\[Link\]](#)
2. **Jiawei Lian**, Xiaofei Wang, Yuru Su, Mingyang Ma, Shaohui Mei. "Contextual Adversarial Attack against Aerial Detection in the Physical World." *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2023. [\[Link\]](#)

## China Patents

- |            |   |                                                                                                                                                  |
|------------|---|--------------------------------------------------------------------------------------------------------------------------------------------------|
| 16/05/2023 | ■ | A method and device for generating adversarial camouflage patterns against object detection. (Patent acceptance number: 202310547602.8)          |
| 16/12/2022 | ■ | A method and device for physical attack-oriented self-adaptive adversarial patch generation. (Patent acceptance number: 202211617496.8)          |
| 15/12/2022 | ■ | A method and device for generating adversarial patches for unobstructed physical adversarial attacks. (Patent acceptance number: 202211612346.8) |
| 05/06/2021 | ■ | A fast surface defect detection method based on the lightweight convolutional neural network. (Patent acceptance number: 202110627893.2)         |

## Miscellaneous Experience

### Honours and Awards

- |           |   |                                                                                                                                 |
|-----------|---|---------------------------------------------------------------------------------------------------------------------------------|
| 2024-2026 | ■ | <b>Joint PhD Program in Dual Degrees between Northwestern Polytechnical University and The Hong Kong Polytechnic University</b> |
| 2023-2025 | ■ | <b>Outstanding Graduate Student</b>                                                                                             |
| 2023      | ■ | <b>Suzhou "Talent Training" Scholarship</b>                                                                                     |
| 2023-2024 | ■ | <b>First Class Academic Scholarship</b>                                                                                         |
| 2023      | ■ | <b>Research featured by New Scientist, a leading UK science and technology media</b>                                            |

### Involved Projects

- |                 |   |                                                                                                                                                                                                     |
|-----------------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 03/2024-Present | ■ | Research on exploring the adversarial vulnerability of large language models.                                                                                                                       |
| 09/2022-02/2024 | ■ | Research on the robustness of visual perception based on deep learning models.                                                                                                                      |
| 12/2021-08/2022 | ■ | Research on background adversarial perturbations for physical attack. <b>Relevant research achievements received an interview invitation from the British authoritative magazine New Scientist.</b> |

### Certifications

- |         |   |                                                                                                                   |
|---------|---|-------------------------------------------------------------------------------------------------------------------|
| 04/2022 | ■ | <b>IELTS: Overall band score 7.0 (Listening 7.5, Reading 7.5, Writing 6.5, Speaking 6.0)</b>                      |
| 07/2017 | ■ | <b>Fitness Coach Licentiate (FCL): Awarded by The Ministry of Human Resource and Social Security, P.R. China.</b> |

### Internship

- |                 |   |                                                             |
|-----------------|---|-------------------------------------------------------------|
| 07/2021-09/2021 | ■ | Internship at Xi'an Automatic Control Research Institute.   |
| 07/2020-09/2020 | ■ | Internship at Xi'an Institute of Modern Control Technology. |